

UNIT 05

STATE OF MATTER (II): LIQUIDS



KINETIC MOLECULAR THEORY OF LIQUID:**LIQUID STATE:**

A state of matter having definite volume but no definite shape with particles that are closely attracted particles with short range of attractive forces.

KINETIC MOLECULAR INTERPRETATION OF LIQUIDS:

The behavior of liquids in terms of kinetic molecular theory can be summarized as:

1. Liquid molecules are close to each other having small spaces between them.
2. Liquid molecules are in constant motion but their movement is restricted, and they perform translational motion.
3. There are weak attractive forces present among liquid molecules.
4. Liquids do not show an appreciable increase in volume by heating.

Properties of Liquids:

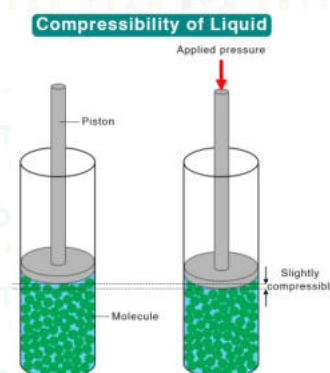
There are some simple properties of liquid e.g. diffusion, compression, expansion, motion of molecules, spaces between them, intermolecular forces, and kinetic energy based on Kinetic Molecular Theory. We can use the Kinetic Molecular Theory to account for the simple properties of liquids:

(i) Diffusion:

The diffusion in liquids takes place because the molecules move from one place to another due to K.E. The restricted movement of the molecules reduces the rate of diffusion e.g. a drop of ink when added to water diffuses slowly due to relatively small empty spaces between the molecules. The diffusion between closely packed molecules of liquids is slow due to less collision between them.

(ii) Compression (effect of Pressure):

A liquid cannot be compressed significantly by increasing the pressure because the molecules are already in close contact with one another e.g. an increase of pressure from one to two atmospheres reduces the volume of water to 0.0045 % which is negligible. However, the same pressure reduces the volume of a gas up to 50%.

**(iii) Expansion (effect of temperature):**

The liquids expand on heating because the intermolecular forces between them decrease. Moreover, the increase of temperature increases the effective collisions between the molecules. If the temperature is decreased contraction of molecules takes place. This property is useful for making thermometers, e.g. mercury thermometer in it, if the temperature rises, the mercury expands in the capillary tube.

(iv) Motion of molecules:

The molecules move with lesser speed due to larger forces of attraction among them. As a result they have lesser kinetic energy. However the kinetic energy increases with the increase of temperature.

(v) Spaces between them:

The molecules forming the liquid states are fairly close to each other. There is very little space between them. As a result, the number of collisions among the molecules are moderate. Therefore, the average kinetic energy is also moderate.



INTERMOLECULAR FORCES (Van der Waal's Forces):

The attractive forces existing between the individual particles of a substance are called intermolecular forces. The physical properties of liquids such as boiling point, vapour pressure, surface tension, viscosity and heat of vaporization depend upon the strength of intermolecular attractive forces.

These forces are electrostatic in nature & known as "Physical bonds or secondary forces."

Main types:

1. Dipole-dipole interaction
2. London dispersion forces
3. Hydrogen bonding
4. Ion-dipole forces
5. Induced dipole forces

DIPOLE-DIPOLE INTERACTION:

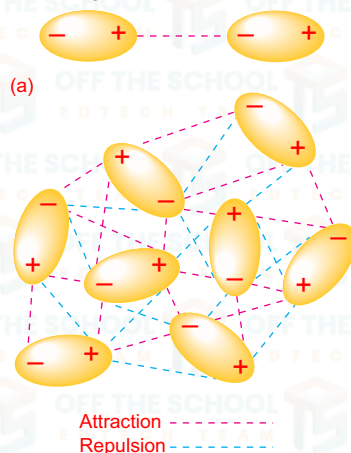
The electrostatic force of attraction present between the positive end of one polar molecule & the negative end of another polar molecule.

EXPLANATION:

- Polar liquid molecules have partial positive & partial negative charges due to electronegativity differences on their opposite ends (poles).
- Example: In iodine mono chloride (Icl), chlorine possesses a partial negative charge while iodine is possessing a partial positive charge. The partial negative terminal of chlorine is attracted to the partial positive terminal of iodine creating dipole-dipole interaction.
- The strength of these forces depends upon the difference in electronegativity between two atoms in the polar molecule.

LONDON DISPERSION FORCES:

"The attractive forces that develop between instantaneous & induce a dipole."

**Instantaneous dipole:**

When one atom's electron cloud gets pushed around by the presence of another atom, it causes the first atom to develop a temporary polarization.

Induce dipole:

The dipole produced by the instantaneous dipole, distorts the electron density of a nearer atom.

Explanation:

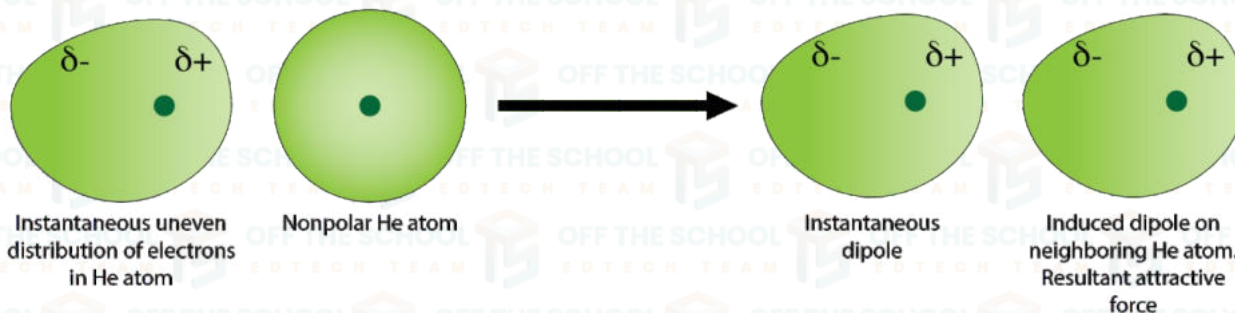
- These forces are reported by Fritz London & these are the attractive forces among non-polar molecules that temporarily become polar.

Example:

Gasoline & benzene are non-polar but there exist London dispersion forces in both of them when existing in a liquid state at room temperature.

- When non-polar substances (H_2 , Cl_2 , F_2 , CH_4 , He, Ne, Ar) are liquefied by lowering the temperature. The interaction between molecules is developed due to instantaneous & induced dipole.





The strength of London dispersion forces depends upon:

1. Size of Atom or Molecule: A larger size means a greater number of electrons. Hence more distortion of electrons possibly increases London forces.

2. Atoms in molecules: The greater the number of atoms in non-polar molecules the more the number of electrons & distortion of electrons increases London forces.

HYDROGEN BONDING (H-Bonding):

When hydrogen (H) is covalently bonded to a highly electronegative atom X (O, N, F), the shared electron pair is pulled so close to X this is hydrogen bonding.

Definition:

"The electrostatic attraction between an H atom covalently bonded to a highly electronegative atom X and a lone pair of electrons of X in another molecule, is called Hydrogen Bonding."

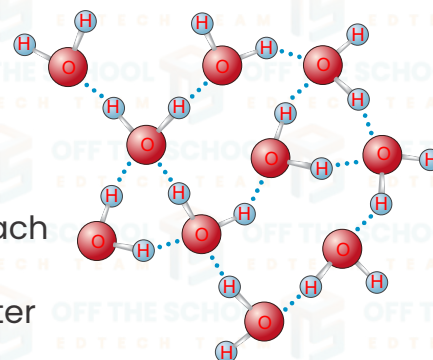
Hydrogen bond is represented by a dashed or dotted line (.....) & known as protonic bridge.

EXAMPLE OF HYDROGEN-BONDED COMPOUND:

Water, H_2O :

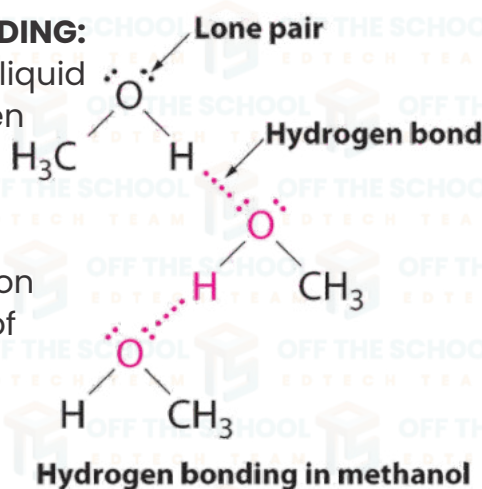
In H_2O molecule, two hydrogen atoms are covalently bonded to the highly electronegative O atom. Here each H atom can hydrogen bond to the O atom of another molecule, thus forming large chains or clusters of water molecules.

Each O atom still has an unshared electron pair which leads to hydrogen bonding with other water molecules. Thus, liquid water, in fact, is made of clusters of a large number of molecules.



APPLICATIONS & IMPORTANCE OF HYDROGEN BONDING:

- Water (H_2O) a necessity of living being exist in the liquid state rather than the gaseous state due to hydrogen bonding while H_2S (a hydride of the second member of the oxygen family) exist in the gaseous state due to the absence of hydrogen bonding.
- The cleaning action of soap & detergent is based on hydrogen bond formation between the polar part of detergent & soap with water molecules.
- Paints are viscous materials but easily form a smooth layer on the surface, the adhesive action of paint is due to hydrogen bonding.
- Fabrics of cloth are made up of fibers such as silk, polyester & nylon. The rigidity & tensile effect in threads is developed by hydrogen bonding.



PROPERTIES OF WATER DUE TO HYDROGEN BONDING!

Water possesses some unique properties due to hydrogen bonding:

1) HIGH SPECIFIC HEAT:

Specific heat is the amount of heat energy required to raise the temperature of one gram of any substance by 1°C .

- Due to hydrogen bonding in water, it does not warm up & cools down rapidly. Thus this property plays an important role in moderating the temperature of the Earth's surface.
- In coastal areas, heat from the sun is absorbed & stored in the daytime and will be available at night. Therefore, temperature remains moderate throughout the day.
- In desert areas, the temperature shoots up in day time but falls steeply at night due to the low specific heat of rocks & sand.

2) HIGH BOILING POINT:

The temperature where the vapor pressure of the liquid becomes equal to the external atmospheric pressure.

- Due to hydrogen bonding in water, it has a high boiling point of 100°C compared to other liquids.
- In water two polar hydrogens & one polar oxygen atom with two non-bonded electron pair is present. Therefore, water can make four hydrogen bonds three-dimensional space, hence extra energy is required to break these secondary bonds.
- The boiling point of HF is lower than H_2O although fluorine is more electronegative than oxygen because in HF molecule the number of hydrogen bonds is lesser than in water due to its linear geometry.

3) HIGH DENSITY:

The mass per unit volume of any substance.

- Due to hydrogen bonding in water, molecules of water are strongly attracted occupy less volume. As density is inversely related to volume, therefore the density of water is relatively high. At 20°C density of water is $1\text{g}/\text{cm}^3$.
- Marine life survives in water due to its high density.

4) HIGH VISCOSITY:

The measure of internal resistance of a liquid to flow.

- Due to hydrogen bonding in water, the viscosity of water is relatively high.
- Due to three-dimensional hydrogen bonding in water molecules, molecules are quite close to each other causing high resistance in flow increases viscosity.

5) HIGH SURFACE TENSION:

The force acting at the right angle to the unit length of the surface of the liquid.

- Due to hydrogen bonding in water, the surface tension of water is relatively high.
- High surface tension in water is actually due to high cohesive forces due to high hydrogen bonding.

6) ANOMALOUS BEHAVIOR:

Water does not expand between 0°C to 4°C but it contracts & expands above 4°C & possesses maximum density at 4°C .

• Density Peak at 4°C :

Water is densest at 4°C means it has maximum density at 4°C .



• Unusual Expansion when Freezing:

Below 4°C , water expands when freezing.

Hydrogen bonds break, molecules arrange hexagonally, creating space.

Volume increases by up to 10%.

• Dynamic Hydrogen Bonds:

Above 4°C , thermal energy disrupts hydrogen bonds.

Bonds break and reform easily.

Allows water molecules to interact freely.

These properties arise from the unique behavior of water molecules and their ability to form and break hydrogen bonds.

TYPICAL PROPERTIES OF LIQUID:**EVAPORATION:**

"The process of changing of a liquid into a gas phase is called evaporation."

Explanation:

It is a surface phenomenon & change of state takes place only at the surface by getting energy from the surrounding.

Based on the rate of evaporation, liquids are classified into two types:

Volatile liquids:

The liquids which have high vapor pressure & evaporate readily.

Example: Ethanol, Acetone, Gasoline etc.

Non-volatile liquids:

The liquids which have low vapor pressure & do not evaporate readily.

Example: Mercury, Mobile oil, Grease

Point to be noted:

Evaporation is endothermic (cooling) process, upon evaporation high-energy molecules leave the surface of the liquid. Therefore, the average kinetic energy of the remaining molecules decreases & temperature of the liquid falls.

Example:

Acetone (a volatile liquid) feels cool on the palm as it absorbs energy from the palm & evaporates readily.

Factors affecting:

i. Temperature: At high temperature, rate of evaporation is high because at high temperature kinetic energy of the molecules increases so high that they overcome the intermolecular forces and evaporate rapidly.

ii. Surface area: Evaporation is a surface phenomenon so, greater is surface area, greater will be evaporation and vice versa.

iii. Intermolecular forces: If intermolecular forces are stronger, molecules face difficulty in evaporation.

**Vapour pressure:**

"The pressure exerted by the vapours of a liquid at equilibrium with the liquid at a particular temperature is called vapour pressure of a liquid."

Explanation:

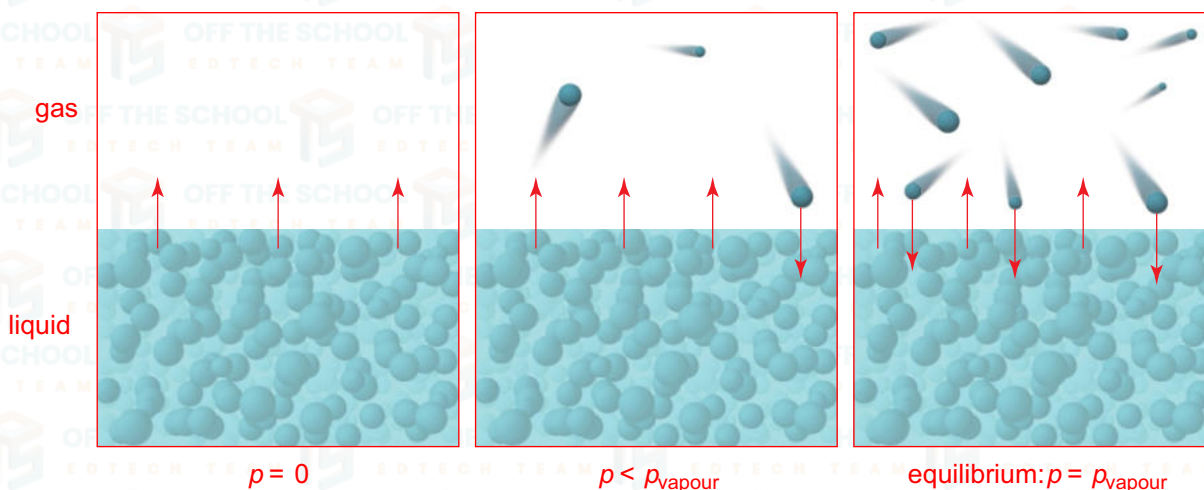
If the liquid is placed in a closed vessel, the molecules with high kinetic energies



escape into space above the liquid. As the number of molecules in the gas phase increases, some of them strike the liquid surface and are recaptured (condensation). A stage comes when the number of molecules escaping from the liquid is equal to the number of molecules returning to the liquid. In other words, the rate of evaporation exactly equals the rate of condensation. Thus, a dynamic equilibrium is established between the liquid and the vapour at the given temperature.



As the partial pressure of a substance in the gas phase above a liquid in a closed container increases, the difference between the number of molecules escaping into the gas phase (represented by the upward red arrows) and the number re-entering the liquid phase (represented by the downward red arrows) decreases until, at the vapour pressure of the substance, equilibrium is established



Factors affecting:

i. Nature of liquid: Vapour pressure depends upon the nature of liquid. Polar liquids have low vapour pressure than non-polar liquids at the same temperature.

ii. Size of molecules: Small sized molecules can easily evaporate than big sized molecules hence, small sized molecular liquids exert more vapour pressure.

iii. Temperature: At high temperature, vapour pressure is higher than at low temperature. Vaporize and exert pressure.

BOILING POINT:

The normal boiling point of a liquid is the temperature at which the vapor pressure is equal to 760 torr or 1 atm.

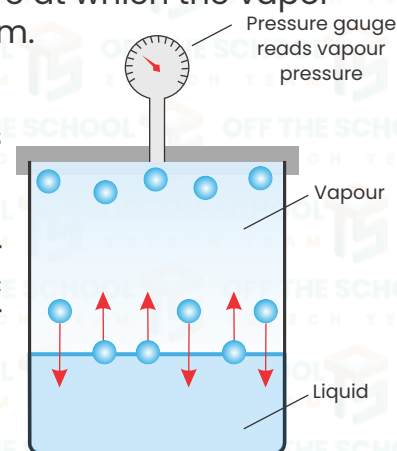
ALTITUDES:

- The boiling point decreases with altitudes, as atmospheric pressure decreases.

Example:

At sea level, atmospheric pressure is 760 torr & water boils at 100°C but in mountainous regions atmospheric pressure is low (example 700 torr) So water

In Pakistan, at higher altitudes like those found in Muree or Nathia Gali, water boils at lower temperatures due to the reduced atmospheric pressure. For instance, in these hill stations, where elevations are significant, water boils at a temperature lower than the standard 100°C. This reduced boiling point makes cooking take much longer, and harmful bacteria may not be completely killed because the water doesn't reach the high temperatures necessary for thorough cooking. Appliances called pressure cookers are sometimes used. They retain some of the water vapor, thus increasing the pressure inside the cooker and increasing the boiling point.



boils below 100°C.

- At a normal boiling point (B.P at 760 torr i.e. 100°C) food will be cooked easily but at higher altitudes pressure is low which lowered the boiling temperature & food takes a large time to cook.

VISCOSITY:

The measure of internal resistance of a liquid to flow.

Explanation:

- The decrease in the flow of liquids due to internal resistance among the layers of molecules.
- Every layer offers some friction to a neighboring layer which tends to flow over it.

Units:

The viscosity is expressed in poise or centipoise; the S.I. unit is $\frac{\text{N.s}}{\text{m}^2}$

Factors affecting viscosity:

1- Intermolecular Forces

Polar molecules are more viscous than non-polar molecules. The stronger the Intermolecular forces, the higher will be viscosity.

Example: Water is more viscous than acetone because of the presence of a stronger hydrogen bond while acetone possesses weaker London dispersion forces.

2- Temperature:

The viscosity of liquid decreases with the rise in temperature due to the increased kinetic energy of molecules.

Example: Vegetable oil becomes less viscous while cooking, and Honey becomes thin on heating.

3- Molecular Size:

The larger the size of molecules, the greater will be viscosity as molecular chain tangled with each other creates more friction.

For example honey is viscous because of its large molecular size & hydrogen bonding similarly paraffin oil & motor oil are viscous because they are long chain hydrocarbons.



SURFACE TENSION:

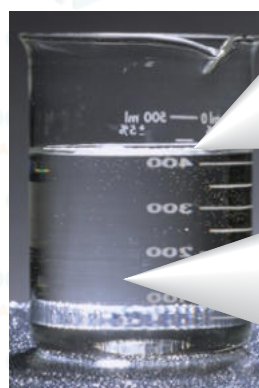
The force acting at the right angle on the unit length of the surface of the liquid.

OR

Surface tension is the resistance of a liquid to spread out and increase its surface area.

EXPLANATION:

- To understand the phenomenon of surface tension, consider two molecules of the liquid, one is present inside the liquid this molecule is equally attracted in all direction.
- Another molecule is present at the surface of the liquid this molecule experienced a downwards force due to no forces of attraction from the upper side.



molecule is pulled away from surface

no net 'pull' in any direction

- Thus, the molecules present at the upper surface bend downwards due to unbalanced force, to attain the smallest surface area to volume ratio.

Example:

Liquid tends to minimize the surface area & attains a spherical shape. Therefore, falling drop of water is spherical due to the presence of unbalanced cohesive forces between the surface & inner molecules. As inner molecules are pulled surface molecules results in a decrease in surface area to volume ratio.

Units:

The surface tension is expressed in newton per meter.

FACTORS AFFECTING EVAPORATION**1- Temperature:**

Surface tension varies inversely with temperature because if the temperature is increased the kinetic energy of molecules increases. Thus, the force of attraction between them decreases.

Therefore, surface tension decreases if temperature increases.

2- Intermolecular Forces of Attraction:

Surface tension is directly related to the intermolecular forces of attraction. If intermolecular forces of attraction are high surface tension increases.

Example: Water has more surface tension than other liquids because of the presence of stronger hydrogen bonds.

CAPILLARY ACTION:

The rise or upwards flow of liquid in the capillary tube.

Explanation:

- Capillary action is due to the surface tension because the lower surface area of a capillary tube is in a curve form & area to volume ratio of a liquid is lesser in a curve shape. Thus liquid rises in a capillary tube because of surface tension to reduce the surface area to volume ratio of liquid.
- The flow of liquid in the capillary tube is due to the combined effect of cohesive & adhesive forces.

COHESIVE FORCES

The force created among the molecules due to inter-attraction between them is called cohesive force.

ADHESIVE FORCES

The force created between the molecules & walls of a container is called adhesive force.



Surface tension causes rainwater to bead on leaves.

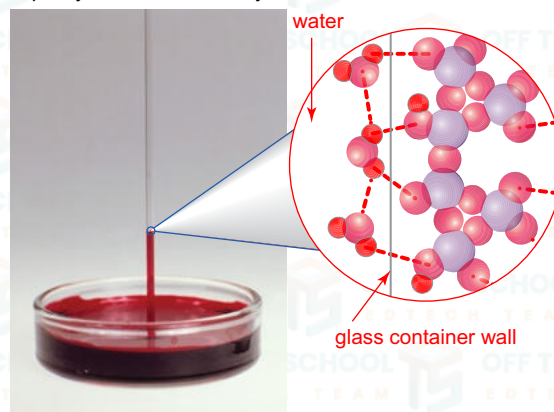


The surface tension of water supports this water strider. The nonpolar surfaces of its feet also help to repel the water.

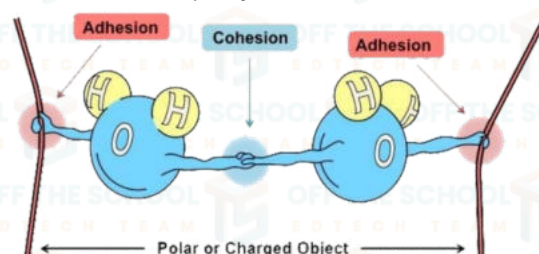


Intermolecular forces acting on a molecule in the surface layer of a liquid and in the interior region of the liquid.

Water rises inside a small-diameter tube cause of capillary action. The water in this photo has been coloured to show the effect of capillary action more clearly.



Blood being drawn from a finger into a capillary tube.



ENERGETIC OF PHASE CHANGE:

The transformation of a substance from one physical state to another is accomplished by the absorption or evolution of heat energy.

MOLAR HEAT OF FUSION (ΔH_{fusion}):

The amount of heat required to convert one mole of a solid completely into liquid at its melting point.

Example: The molar heat of fusion of ice is given as;

**MOLAR HEAT OF VAPORIZATION (ΔH_v):**

The amount of heat required to convert one mole of a liquid into its vapor at its boiling point.

Example: The molar heat of vaporization of water is given as,

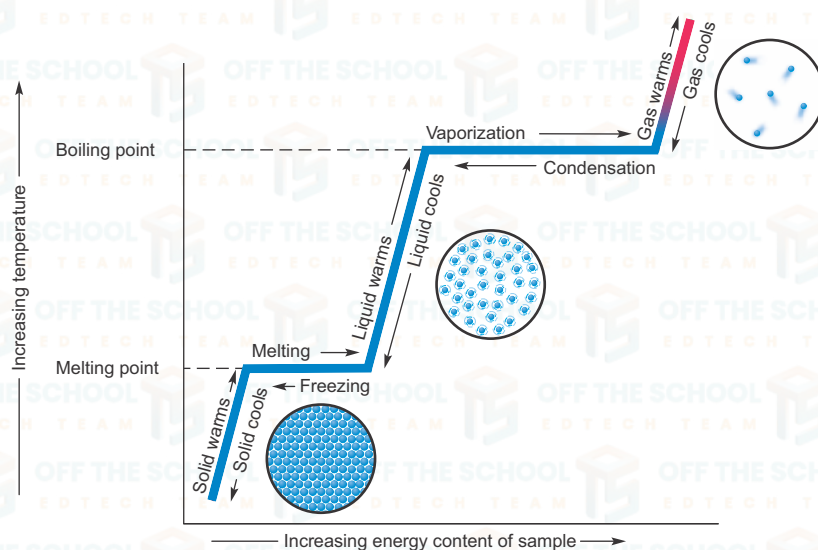


The heat of vaporization is greater than the heat of fusion, as high energy is required to bring a notable change in intermolecular distance & overcome potential energy.

MOLAR HEAT OF SUBLIMATION (ΔH_s):

The amount of heat absorbs when one mole of a solid is directly converted into vapor without going through the liquid phase.

Example: The molar heat of sublimation of dry ice (solid CO_2) given as,

**LIQUID CRYSTALS:**

The state of matter between liquid & solid.

OR

A substance that flows like a liquid but has a molecular arrangement similar to that of a crystalline solid.

Discovery & explanation of liquid crystals:

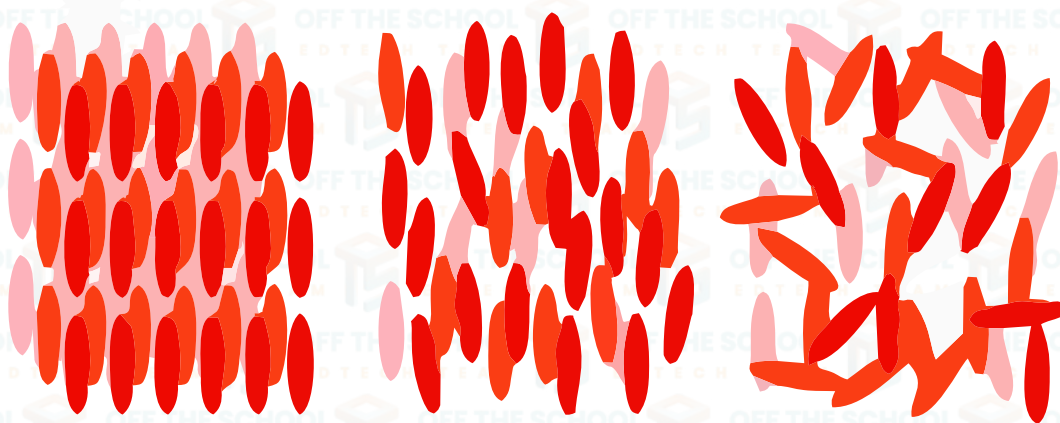
- Liquid crystals are discovered by Fredrich Reintzer (1888).
- Working on the physicochemical properties of derivatives of cholesterol, He observed that cholesteryl-benzoate did not melt as other solid does.
- At 145.5°C it first melts to form a cloudy liquid & then at 178.5°C it melted again & cloudy liquid becomes clear. The phenomenon observed is reversible.
- Liquid crystals have optical properties like solids but possess anisotropy.



- Liquid crystals are usually organic, but sometimes they are a mixture of organic & inorganic materials.
- Liquid crystals are both natural & synthesized (LCD).

Uses of liquid crystals:

- They are commonly used in daily life such as liquid crystal displays (LCDs) used as the screen of mobile, smart watches, tablets, computers, pH meters, etc.
- They are used as a temperature sensors in thermometers & devices to measure temperature.
- They are used to detect faulty connections in microelectronic circuit boards.
- They are used to detect blockage of veins, arteries & tumors.
- They are used in chromatographic techniques as a solvent for the separation of many components.
- They are used in hydraulic machines due to their high viscosities.



Scientific Reason(s):

Why B.P of H_2O is greater than HF instead of the fact that F is more electronegative than O?

The B.P. of the compound depends upon the strength of the hydrogen bond present in it H_2O is angular in shape thus hydrogen bonds are present in a network pattern while in HF it is linear so, hydrogen bonds exist linearly. Therefore, the number of H.B. in H_2O is greater than that in HF. Thus, the B.P. of water is high.

Why water contracts on heating and expand on cooling which is anomalous to other liquids?

Water contracts on heating and expand on cooling because on cooling the H.B. present between the molecules of water dissociates and molecules of water are arranged in a tetrahedral fashion thus, upon the cooling distance between the molecules of water increases hence water expands on cooling.

Why water has a maximum density at 4°C ?

Water has maximum density at 4°C because on cooling below 4°C H.B. breaks & expansion occurs on heating above 4°C due to a rise in temperature K.E. of molecules increases which causes breakage of H.B. hence again expansion occurs. Thus, at 4°C water has maximum numbers of H.B.



Why Boiling point and viscosity of Sulphuric acid is high?

Hydrogen bonding is present between the molecules of Sulphuric acid as a result of which forces of attraction between the molecules of Sulphuric acid increase which result in enhancement of B.P. and viscosity of Sulphuric acid.

Why B.P. of liquid on Mount Everest is less than that on sea level?

The B.P. depends on the external atmospheric pressure and it is directly proportional to it. The external atmospheric pressure on the sea level is more than that at altitudes (such as Mount Everest) Therefore, B.P. decreases i.e. liquid boils earlier than its normal B.P.

Why food cooks rapidly in a pressure cooker?

The pressure cooker itself develops its internal pressure by preserving the steam to leave from the container as a result of which the vapor pressure of liquid increases hence B.P. also increases than normal B.P. Hence, food utilizes more amount of heat that facilitates the cooking process.

Why Pressure cooker is used for cooking food in Mountainous regions?

The pressure cooker itself develops its internal pressure by preserving the steam to leave from the container as a result of which the vapor pressure of liquid increases hence B.P. also increases, which is required in a mountainous region.

Why Glycerin is distilled under reduced pressure?

Glycerin dissociates before its boiling point so it cannot be purified by distillation at ordinary pressure. As B.P. is directly proportional to atmospheric pressure, therefore the pressure is reduced for its purification so that glycerin boils earlier than its dissociation temperature and is purified without dissociation.

Science may set limits to knowledge, but should not set limits to imaginations.

